QPAD: Calibration-Set Composition Bias in INT8 Face Presentation Attack Detection

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Nature of project: Test whether INT8 calibration-set composition creates attack-specific regressions in face presentation attack detection.

***Abstract*—We study whether the class mixture of the post-training INT8 calibration set changes how often a face presentation attack detector (DeepPixBiS) accepts print or replay attacks on OULU-NPU Protocol II. This checkpoint reports the data engineering pipeline, the dataset audit, exploratory analysis and the FP32 baseline with a development-set operating threshold. Blue text marks changes in scope since the project proposal. Code, notebooks and the scripts that generate every number in this report: https://github.com/AnanyaTyagi/qpad-int8-face-pad (tag `checkpoint1`).**

I. Problem Statement and Motivation

A face presentation attack detection (PAD) system decides whether a camera sees a live person (*bona fide*) or a presentation attack instrument (PAI) such as a printed photograph or a video replayed on a screen. A missed attack is a security failure: an impostor holding a photo or a phone is accepted as the enrolled user. Errors are therefore reported separately as APCER (attacks accepted) and BPCER (bona fide presentations rejected) following ISO/IEC 30107-3 [1].

Deployed PAD models are often quantized to INT8 for speed [2], [3]. Post-training INT8 quantization keeps the trained weights but selects per-tensor value ranges from a small *calibration set*. If that set over- or under-represents an attack type, the chosen ranges might clip exactly the activations that distinguish that attack, raising its APCER even though no weight changes.

Research question. Does the composition of the INT8 calibration set cause attack-specific degradation—in print or replay APCER—relative to an INT8 engine calibrated on the natural class mixture, after every engine receives its own development-set threshold? The primary comparison is *unusual-calibration INT8 versus natural-calibration INT8 of the same nested seed*, which isolates calibration composition from ordinary quantization error. A meaningful effect is pre-registered as an attack-specific APCER increase of 5 percentage points; a null result with tight confidence intervals is an acceptable, informative outcome.

II. Data Engineering

A. Dataset origin, access and characteristics

We use OULU-NPU [4], a mobile face PAD database of 55 subjects recorded with the front cameras of six smartphones in three sessions with different illumination and background, with print and video-replay attacks produced by two printers and two displays [4]. OULU-NPU is distributed for research use under an end-user licence agreement; our copy was provided by the course instructor on request for this course project. The data are kept in private storage and are not redistributed, and no face images appear in this report.

TABLE I
Protocol II videos per split and class (from the audit manifest).

Split	Bona fide	Print	Replay	Total
Train	360	360	360	1080
Development	270	270	270	810
Test	360	360	360	1080

Our archive audit found 4950 video files in the three split archives, of which 2970 belong to Protocol II. Protocol II videos have resolution 1080x1920 (2970) (width × height, number of videos), 41–151 (median 146) frames, a median of 29.84 frames/s and duration 3.10–5.06 s (median 4.97 s). In Protocol II every subject contributes 54 videos: 6 phones × 3 sessions for each of bona fide, print and replay. Labels are bona fide versus attack, with the attack type (print or replay) and the subject identity. Each video has a same-name text file with per-frame eye coordinates (`frame, x1, y1, x2, y2`).

B. Protocol II splits and labels

Filenames encode `phone_session_subject_PAI`. PAI code 1 is bona fide, 2–3 print and 4–5 replay. Protocol II trains and selects thresholds on Printer 1/Display 1 attacks (codes 2, 4) and tests on unseen Printer 2/Display 2 attacks (codes 3, 5); subjects 1–20, 21–35 and 36–55 form disjoint train, development and test splits (Table I). The development split is used only for model selection and threshold choice. The test split is scored only after the threshold is frozen; a lock file records the threshold's hash and refuses any later test evaluation with a different threshold.

C. Schema, quality checks and cleaning

One manifest row per video records split, subject, phone, session, PAI code, class, frame count, resolution, FPS, duration, annotation statistics, decode status and file size. Overall audit status: **PASS**. The audit verifies: the 2,970 expected videos exist with the expected split/class counts; the subject/PAI membership rule agrees with the official Protocol II

list files; no subject appears in two splits; there are no duplicate or unparseable identifiers; every video decodes completely with FFmpeg; every annotation file is well formed; and every video, apart from the two documented test exceptions below, has at least 20 valid annotated frames. Annotation rows whose coordinates are all zero (missing eye detections) are treated as invalid frames, not imputed. Videos failing at least one per-video check: 3 (6_1_12_1, 3_1_54_5, 5_1_54_3). All 2,970 videos are kept, with a documented rule fixed before any training or test scoring: one training video has a single damaged MJPEG frame, which is excluded from frame selection; two test attack videos have only 2 and 19 frames with valid eye coordinates and are scored on all of those frames instead of 20. We keep rather than drop them because removing attacks that are hard to annotate could make APCER look better than it is.

D. Preprocessing

For each video we select 20 evenly spaced frames among frames with valid eye annotations and save their indices. Using the supplied eye coordinates (no separate detector), an exact similarity transform places the two eyes at fixed positions of a 224×224 crop; frames are converted to RGB and normalized with ImageNet statistics at load time, identically for every backend. For calibration, five evenly spaced frames of each training video are saved. No feature-reduction method such as PCA is used, because DeepPixBiS learns image features directly from pixels. Processed crops are stored in verified, compressed shards in private storage.

E. Shortcomings and limitations

Protocol II covers two printers and two displays only; subjects are few (20 test subjects), so test videos are clustered by subject; uncertainty estimates in the INT8 analysis will therefore resample subjects, not videos. Because test attacks use equipment unseen in training (Printer 2, Display 2), a "print" or "replay" effect is also an effect on that specific instrument. Each calibration set is 120 videos $\times$ 5 frames; frames of one video are correlated, so the effective calibration sample is closer to 120 than to 600. Eye annotations are used as supplied. Conclusions are limited to OULU-NPU Protocol II.

III. Exploratory Data Analysis

Fig. 1 shows class balance: attacks outnumber bona fide videos two to one, so a single accuracy would hide attack-specific errors; we always report per-class rates. Fig. 2 (histogram) and Fig. 3 (box plot) are univariate views of video length and crop sharpness, and Fig. 4 a bivariate view of crop brightness versus sharpness per video. Pixel statistics use train and development videos only. Within every split the three classes are exactly balanced (Table I), and every phone and session contributes the same number of videos per class, so no device or recording condition dominates. On the aligned crops of the 1,890 train and development videos (630 per class),

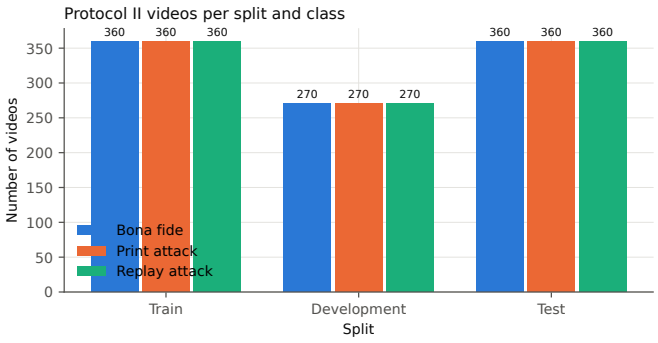


Fig. 1. Class balance per split. Every split has equal bona fide, print and replay counts, so attacks outnumber bona fide videos two to one and APCER and BPCER must be reported separately.

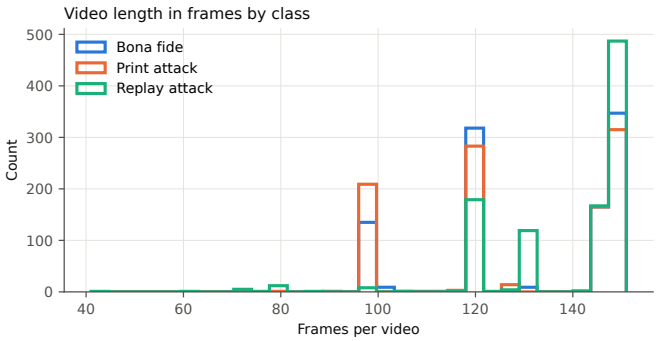


Fig. 2. Univariate: decoded frames per video by class (all Protocol II videos). Video length varies from 41 to 151 frames, so we score a fixed number of frames per video rather than all frames.

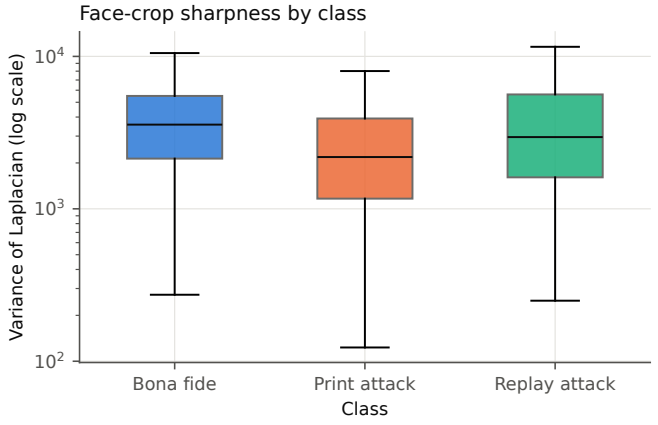


Fig. 3. Univariate: per-video sharpness (variance of Laplacian) of aligned crops by class (train + development). Print crops are the least sharp, consistent with detail lost when a face is printed and re-captured.

attacks are brighter than bona fide presentations (median per-video grey level 109.7 for bona fide, 130.8 for print and 144.7 for replay), and print attacks are less sharp (median variance of Laplacian 2186, versus 3568 for bona fide and 2955 for replay; Fig. 3), consistent with the loss of high-frequency detail when a face is printed and re-captured. The three classes nevertheless

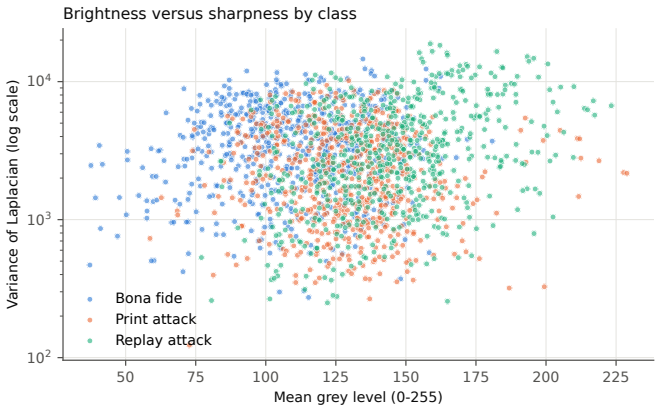


Fig. 4. Bivariate: brightness versus sharpness, one point per video, coloured by class (train + development). The classes overlap heavily, so these global statistics cannot separate attacks on their own.

TABLE II
FP32 DEEPPIXBIS AT THE FROZEN DEVELOPMENT THRESHOLD.

Metric	Development	Test
Print APCER	2/270 = 0.74%	2/360 = 0.56%
Replay APCER	0/270 = 0.00%	25/360 = 6.94%
APCER (pooled)	2/540 = 0.37%	27/720 = 3.75%
APCER (worst PAI)	2/270 = 0.74%	25/360 = 6.94%
BPCER	1/270 = 0.37%	9/360 = 2.50%
ACER (pooled)	0.37%	3.12%
ACER (worst PAI)	0.56%	4.72%
EER (own scores)	0.37%	2.78%
AUROC	0.9998	0.9963

overlap heavily in the brightness–sharpness plane (Fig. 4), so these global image statistics cannot separate attacks on their own; the detector has to learn finer, local cues, which motivates a pixel-wise model.

IV. BASELINE IMPLEMENTATION AND PRELIMINARY RESULTS

A. Model and training

DeepPixBiS [5] uses the first eight blocks of an ImageNet-pretrained DenseNet-161 [6], a 1×1 convolution with sigmoid giving a 14×14 realness map, and a linear layer giving a binary output; the loss is 0.5 pixel-wise BCE $+\,0.5$ binary BCE. We train once on the training split (Adam, learning rate 10^{-4}, weight decay 10^{-5}, batch 32, horizontal flips), select the epoch with the lowest development loss, and freeze the checkpoint (SHA-256 recorded). Seeds are fixed: seed 42 for training, data-loader shuffling and augmentation; frame selection and the threshold rule are deterministic. The frame score is the mean of the map and the video score the mean of 20 frame scores (2 and 19 for the two documented reduced-frame test videos). Training ran on one NVIDIA A100 GPU and stopped early after 25 epochs; the selected epoch 20 has development loss 0.0532 (Fig. 5). After a short single-epoch spike, the development loss flattens while the training loss keeps falling, which is the point at which early stopping is meant to act.

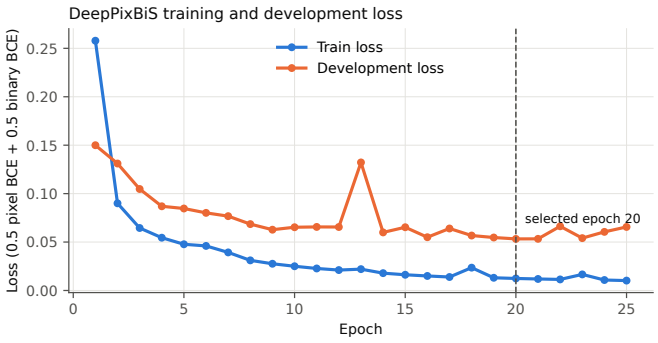


Fig. 5. Training and development loss per epoch; the dashed line marks the selected checkpoint. Development loss stops improving while training loss keeps falling, so later epochs are not used.

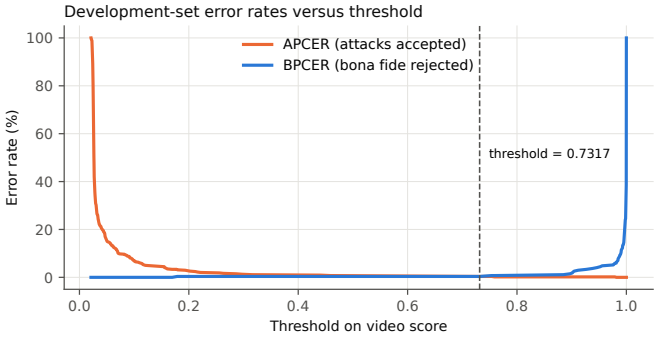


Fig. 6. Development APCER and BPCER versus threshold; the dashed line is the frozen operating point. It sits where the two error rates cross (the EER point), chosen without test data.

B. Threshold analysis and operating point

A video is bona fide when score $\geq \tau$. For PAD, APCER and BPCER play the roles of FAR and FRR. On development scores we enumerate all candidate thresholds, minimise $|\text{APCER} - \text{BPCER}|$, break ties by the lowest ACER and then the higher threshold. The selected $\tau = 0.731701$ gives development APCER 2/540 and BPCER 1/270. This EER operating point balances security and usability without using test data.

C. Results

Table II reports every rate with its numerator and denominator. Test print APCER is 2/360 = 0.56%, replay APCER 25/360 = 6.94%, BPCER 9/360 = 2.50%, ACER 3.12% (worst-PAI convention: 4.72%), EER 2.78% and AUROC 0.9963. Development performance is near perfect (APCER 2/540, BPCER 1/270, AUROC 0.9998), whereas test performance is clearly lower because Protocol II tests on Printer 2 and Display 2, which the model never saw. The drop is almost entirely in replay attacks: print APCER stays below 1%, while replay APCER rises from 0/270 on development to 25/360 = 6.94% on test. George and Marcel report APCER 11.39%,

TABLE III
TEST ERRORS BY PHONE (LEFT) AND BY SESSION (RIGHT) (ERRORS/TOTAL).

Phone	Print APCER	Replay APCER	BPCER
1	0/60	2/60	3/60
2	2/60	13/60	1/60
3	0/60	0/60	2/60
4	0/60	8/60	1/60
5	0/60	2/60	1/60
6	0/60	0/60	1/60

Session	Print APCER	Replay APCER	BPCER
1	1/120	14/120	3/120
2	0/120	8/120	2/120
3	1/120	3/120	4/120

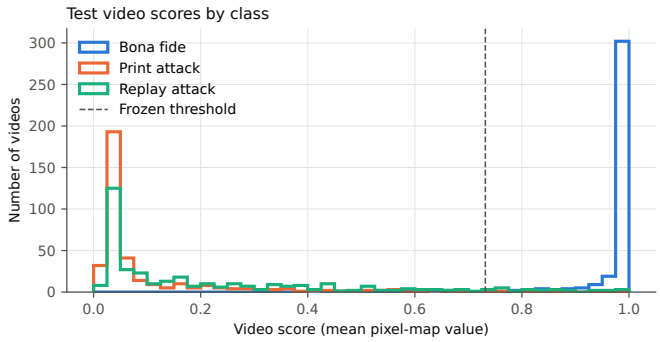


Fig. 7. Test video scores by class with the frozen development threshold (dashed). Attacks to the right of the threshold are accepted; a tail of replay scores crosses it.

BPCER 0.56% and ACER 5.97% on Protocol II, with the same worst-PAI APCER and development-EER threshold [5]. Our reproduction reaches a similar ACER (4.72%) with a lower worst-PAI APCER (25/360 = 6.94%) but a higher BPCER (9/360 = 2.50%). Unlike the original paper, which found print attacks harder than video attacks in most protocols, our errors are dominated by replay. Our recipe differs from theirs in ways that could explain this: we align faces with the supplied eye coordinates instead of MTCNN detection and SDM landmarks, we do not undersample the majority class, we use horizontal flips without colour jitter, and we stop early on development loss instead of training for 50 epochs. We therefore treat the published result as a sanity reference, not a target.

D. Error analysis

Table III breaks test errors down by phone and session for each attack type. Replay attacks account for 25 of the 27 accepted attacks. These errors are concentrated on two phones, phone 2 (13/60) and phone 4 (8/60), while phones 3 and 6 have none, and they decrease from session 1 (14/120) to session 3 (3/120). Bona fide rejections are spread evenly (1–3 per phone, 2–4 per session), so no single device causes them. The test score distribution (Fig. 7) shows the cause: most attack scores lie near 0 and most bona fide scores near 1, but a long tail of replay scores extends past the frozen threshold. Our hypothesis is that the replay cues the model learned from the training display (screen texture, moiré patterns and reflections) look different on the unseen test display, and that these differences are captured more weakly by some phone cameras; per-phone crops of the misclassified replays will be inspected to test this. Replay on unseen displays is therefore the weakest point of the FP32 baseline, and the attack type for which an INT8-induced change in APCER has the most room to appear.

E. Next steps

Export the frozen model to ONNX and TensorRT FP32 (score tolerance 10^{-5} and 10^{-4}), audit calibration scales, and build a six-engine INT8 pilot (natural, no-print, no-replay; two nested seeds) evaluated with both the reused FP32 threshold and engine-specific development thresholds, before deciding on the full 35-engine study.

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